PHY 1221 : Applied Electricity and Magnetism

Section : A

(i) Electrostatics : Electric dipole ; Electric field due to a dipole ; Dipole on external electric field ; Gauss's Law and its applications.

(ii) Capacitors : Parallel plate capacitors with dielectric ; Dielectrics and Gauss's law ; Susceptibility ; Permeability ; and dielectric constant ; Energy stored in an electric field.

(iii) Electric Current : Electron theory of conductivity ; Conductors ; Semiconductors and insulators ; Superconductors ; Current and current density ; Kirchhoff's Law and its applications.

Section-B

(I) Electromagnetic Induction:

Faraday's experiment; Faraday's law; Ampere's law; Motional e.m.f; self and mutual inductance galvanometers- moving coil; Ballistic and deadbeat types.

(II) Thermoelectricity:

Thermal e.m.f; Seabeck; Peltier and Thomson Effects; Laws of addition of addition of thermal e.m.f; Thermoelectric power.

(III) DC and AC Circuits:

D.C. circuits with LR; RC and LCR in series; A.C. circuits with LR; RC; LC; and LCR in series; Power calculation; Power factor improvement.

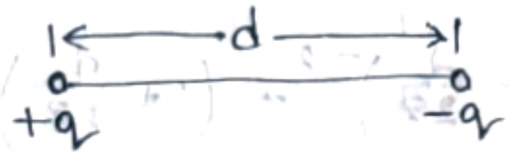
Electrostatics

→ Electric dipole
→ Electric field due to a dipole
→ Dipole on external electric field
→ Gauss's law and it's applications.

Date : 16.11.21 (tues) | class : 01

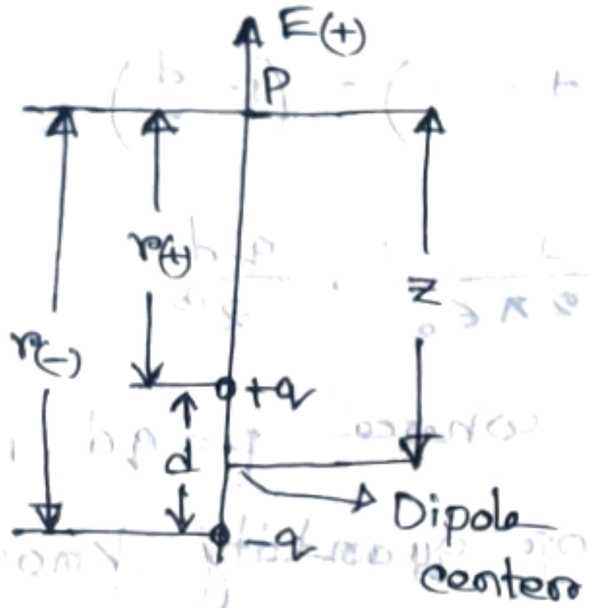
① Electrostatics

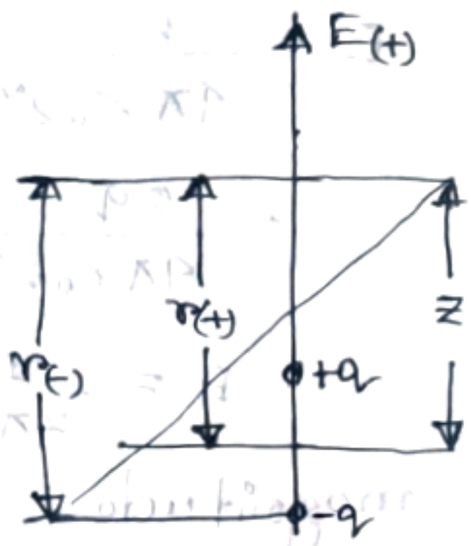
☞ Electric dipole :



01(a) ☞ (Electric dipole moment), $p = qd = q \cdot 2l$

The electric field due to an electric dipole :





$$E = E_{(+)} - E_{(-)}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r_{(+)}^2} - \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r_{(-)}^2}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{\left(z - \frac{1}{2}d\right)^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{\left(z + \frac{1}{2}d\right)^2}$$

$$= \frac{q}{4\pi\epsilon_0}\left[\frac{1}{\left\{z\left(1-\frac{d}{2z}\right)\right\}^2} - \frac{1}{\left\{z\left(1+\frac{d}{2z}\right)\right\}^2}\right]$$

$$= \frac{q}{4\pi\epsilon_0}\left\{\frac{1}{z^2\left(1-\frac{d}{2z}\right)^2} - \frac{1}{z^2\left(1+\frac{d}{2z}\right)^2}\right\}$$

$$= \frac{q}{4\pi\epsilon_0 z^2}\left\{\left(1-\frac{d}{2z}\right)^{-2} - \left(1+\frac{d}{2z}\right)^{-2}\right\}$$

$z >> d$ and $\frac{d}{2z} << 1$.

$$\left[\left(1+\frac{2d}{2z(1)}+\cdots\right) - \left(1-\frac{2d}{2z(1)}+\cdots\right)\right]$$

$$E = \frac{q}{4\pi\epsilon_0 z^2}\left(1+\frac{d}{z}+\cdots\right) - \left(1-\frac{d}{z}\right)$$

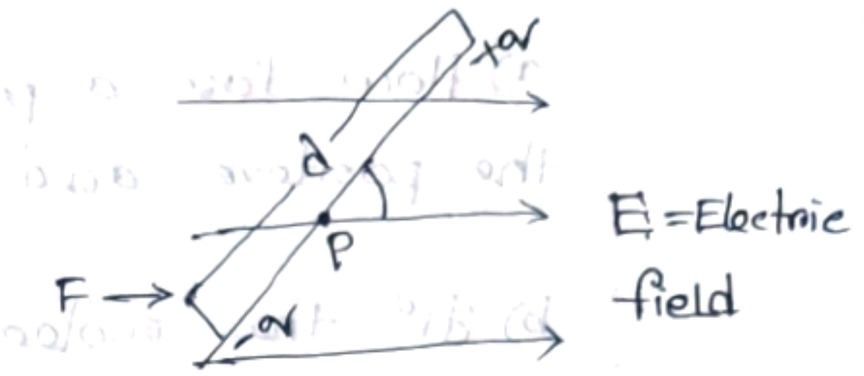
$$= \frac{2q}{4\pi\epsilon_0 z^2}\cdot\frac{2d}{z} = \frac{1}{2\pi\epsilon_0}\cdot\frac{qd}{z^3}$$

$\therefore E = \frac{1}{2\pi\epsilon_0}\cdot\frac{P}{z^3}$ where $P = qd$ is the magnitude of a vector quantity known as the electric dipole moment of the dipole.

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☞ Potential energy of a dipole in a uniform Electric field:

Force, $F = qE$



Torque, $\tau = Fd\sin\theta$

$= qEd\sin\theta$

$= Eqd\sin\theta$

$= EP\sin\theta \quad [\because P = qd]$

$\Rightarrow \boxed{\tau = PE\sin\theta}$

$\therefore \boxed{\vec{\tau} = \vec{P} \times \vec{E}}$

when $\theta = 90^\circ$, then τ is maximum.

$\therefore \tau = PE\sin\theta$

$= PE\sin 90^\circ$

$\therefore \boxed{\tau_{(max)} = PE}$

$\sin\theta = 1$
$\Rightarrow \sin\theta = \sin 90^\circ$
$\because \theta = 90^\circ$

Potential Energy, $W = \int_{90}^{\theta} \tau\, d\theta$

$= \int_{90}^{\theta} PE\sin\theta\, d\theta$

$= PE\,[\cos\theta]_{90}^{\theta}$

$= +PE\cos\theta + PE\cos 90^\circ$

$= +PE\cos\theta + 0$

$\Rightarrow W = +PE\cos\theta = U(\theta)$

when, the value of θ is given, then $W = U(\theta)$.

when, $W = max$, $\theta = ?$

$\cos\theta = 1 \Rightarrow \theta = 0^\circ$

Then, $W = U(\theta) = +PE\cos\theta$

$= +PE\cos 0^\circ$

$\boxed{U(\theta) = +PE}$

Q⇒ A neutral water molecule (H_2O) in its vapor state has an electric dipole moment of magnitude 6.2×10^{-30} Cm.

a) How far apart are the molecules centres of the positive and negative charge?

b) If the molecule is placed in electric field of 1.5×10^{4} N/C. What maximum torque can the field exert on it?

Soln:

a) Given that,
Electric dipole Moment, $P = 6.2\times10^{-30}$ Cm

Molecule charge, $q = 10e$
$= 10\times1.6\times10^{-19}$
$= 16\times10^{-19}$ C

$H_2O = 1\times2+8$
$= 2+8$
$= 10$

We know, $p = qd$

$\Rightarrow d = \frac{p}{q} = \frac{6.2\times10^{-30}\ Cm}{16\times10^{-19}\ C}$

$= 0.38\times10^{-30+19}$

$\therefore d = 0.38\times10^{-11}$ m

b) We know,

$\tau = PE\sin\theta$
$= 6.2\times10^{-30}\times1.5\times10^{4}\times\sin90^\circ$
$\therefore \tau = 9.3\times10^{-26}$ Nm

Ans

where,
$p = 6.2\times10^{-30}$ Cm
$E = 1.5\times10^{4}$ N/C
$\theta = 90^\circ$ [$\because$ Max]

Q5 An electric dipole consists of charges $+2e$ and $-2e$ separated by 0.78 nm. It is in an electric field of strength 3.9×10^6 N/C.
Calculate the magnitude of the torque on the dipole when the dipole is

(i) Parallel

(ii) Perpendicular

(iii) Anti ~~perpen~~ parallel to the electric field.

Soln:

Here, $E = 3.9 \times 10^6$ N/C

and, $p = qd = 2 \times e \times 0.78 \times 10^{-9}$

$= 2 \times 1.6 \times 10^{-19} \times 0.78 \times 10^{-9}$

$\therefore p = 2.496 \times 10^{-28}$ Cm.

$q = 2e = \cdots$
$d = 0.78$ nm
$= 0.78 \times 10^{-9}$ m

(i) When, the dipole ~~moment~~ is parallel,

then, $\theta = 0^\circ$

$\therefore \tau = pE \sin\theta = pE \sin 0^\circ$

$= 2.496 \times 10^{-28} \times 3.9 \times 10^6 \times 0$

$= 0$ Ans

(iii) When, the dipole is perpendicular,

then, $\theta = 90^\circ$

$\therefore \tau = pE = 2.496 \times 10^{-28} \times 3.9 \times 10^6$

$\therefore \tau = 8.98 \times 10^{-22}$ Nm Ans

(iii) When the dipole is anti-parallel,

then, $\theta = 180^\circ$

$\therefore \tau = pE \sin\theta$

$= 2.496 \times 10^{-28} \times 3.4 \times 10^{6} \times \sin 180^\circ$

$= 0 \text{ Nm}$

$\therefore \tau = 0 \text{ Nm Ans}$

Date: 21.11.21

class:03

Electroic flux:

Graussain Surface:

Under necessary condition, gauss law is applying for uniforom symmetrical and distributeional surface, this surface is called gaussian surface.

Characteristics of Graussian surface:

i) The surface is hypothetical not real.

ii) The surface maybe circular, spherical. or any other shape.

iii) The surface must be closed.

iv) Charge is uniformly distributed through the surface.

(E+) 18-01* (a+b)

State and proof gauss's law of electrostatics:

Statement: The total flux of the electric field $\bar{E}$ over any closed surface is equal to $1/\epsilon_0$ times the total net charge enclosed by the surface.

$$\phi = \oint \bar{E} \cdot d\bar{s} = \frac{q}{\epsilon_0} \quad \text{——— (1)} \quad]\ 18\text{-}01(a)$$

Proof:

Consider a single point charge +q located at a point O inside a closed surface, S, fig-1.

Let $d\bar{s}$ be a small area element at a distance r from q.

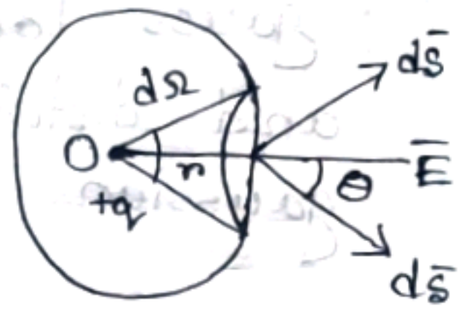


Fig:01

$$\bar{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} \cdot \hat{r}$$

The flux through the area $d\bar{s}$ is given by

$$d\phi = \bar{E} \cdot d\bar{s}$$

$$= E\, ds \cos\theta$$

$$= \left(\frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}\right) \cdot ds \cos\theta$$

$$= \frac{q}{4\pi\epsilon_0} \frac{ds\cos\theta}{r^2}$$

But, $\frac{ds\cos\theta}{r^2} = d\Omega$ = solid angle.

$$\therefore d\phi = \frac{q}{4\pi\epsilon_0} \cdot d\Omega$$

The total flux through the entire closed surface S is given by

$$\phi = \oint d\phi$$

$$= \frac{q}{4\pi\epsilon_0} \oint d\Omega$$

$$= \frac{q}{4\pi\epsilon_0} \cdot 4\pi \quad \left[\because \oint d\Omega = 4\pi\right]$$

$$\therefore \phi = \frac{q}{\epsilon_0} .$$

Differential form of gauss's law:

Let ρ be the charge density, then the total charge within the closed surface enclosing the volume is given by,

$$\text{charge density, } Q = \int \rho \, dv \quad \text{——} (1)$$

We can write integral form of gauss's law as:

$$\oint \bar{E} \, d\bar{s} = \frac{1}{\epsilon_0} \int \rho dv \quad \text{——} (2)$$

By gauss's divergence theorem,

$$\oint \bar{E} \cdot d\bar{s} = \int (\nabla \cdot E) \, dv \quad \text{——} (3)$$

Comparing equation (2) and (3),

$$\int (\nabla \cdot E) \, dv = \frac{1}{\epsilon_0} \int \rho dv \quad \text{——} (4)$$

Since, this true for any volume v, integrands must be equal.

$$\text{Therefore, } \nabla \cdot E = \frac{\rho}{\epsilon_0} .$$

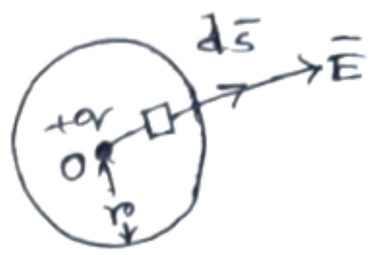
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Ex

Lecture: 04

Deduction of coulomb's Law from gauss's law:

Let us apply gauss's law to an isolated point charge +q lying in vacuum, the gaussian surface is spherical surface of radius r having its centre at the point charge. $\bar{E}$ and $d\bar{s}$ at any point on the surface are directed radially outward angle between $\bar{E}$ and $d\bar{s}$ is zero.



$$\therefore \bar{E} \cdot d\bar{s} = E \cdot ds \cos 0^\circ$$

$$= E ds$$

Applying gauss's law,

$$\oint \bar{E} \cdot d\bar{s} = \oint E\, ds = \frac{q}{\varepsilon_0}$$

E is constant at all point on the surface,

Thus, $$E \oint ds = \frac{q}{\varepsilon_0}$$

$$\Rightarrow E \cdot (4\pi r^2) = \frac{q}{\varepsilon_0} \qquad \left[\because \oint d\bar{s} = 4\pi r^2\right]$$

$$\Rightarrow E = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{r^2}$$

Let us put a second point charge q' on this surface, the force $\bar{F}$ that acts on this charge q' is $F = q'E$

$$= \frac{1}{4\pi\varepsilon_0} \cdot \frac{qq'}{r^2}$$

Question:

Find the Electoric field due to a uniformly charge sphere

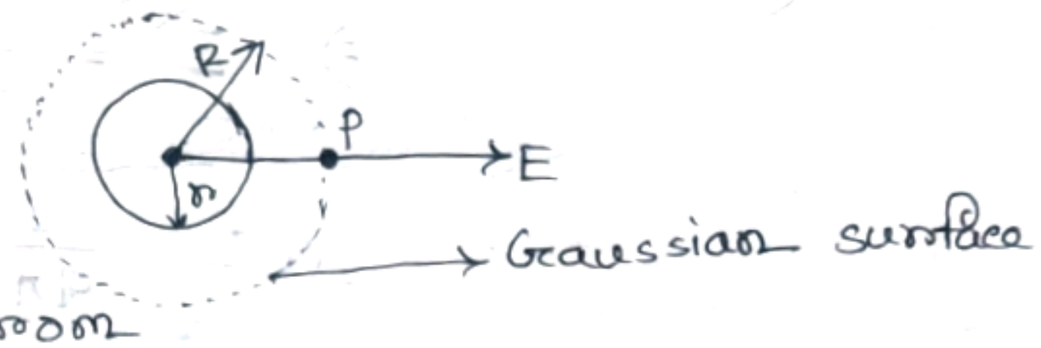
(i) when the point P lies outside the sphere.

(ii) when the point lies on the surface.

(iii) when the point lies inside the sphere.

Soln:

(i)



Let, P is a point at a distance r from the centre O fig (i). We have to find the electoric field $\bar{E}$ at P. Draw a concentric sphere (shown dotted) at radius OP with centre O. This is the gaussian surface at all points of this sphere. The magnitude of the electric field is the same and direction is perpendicular to the surface. angle between $\bar{E}$ and $d\bar{s}$ is zero.

The flux through this surface is given by

$$\oint \bar{E}\, d\bar{s} = \oint E\, ds \cos 0^\circ$$

$$= \oint E\, ds$$

By gauss's law,

$$\oint E ds = \frac{q}{\epsilon_0}$$

Therefore, $\oint E ds = E \oint ds$

$$\Rightarrow \frac{q}{\epsilon_0} = E (4\pi r^2)$$

$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}$$

$$\therefore E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} .$$

(ii) When the point lies on the surface, here $r = R$.

$$\therefore E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{R^2} .$$

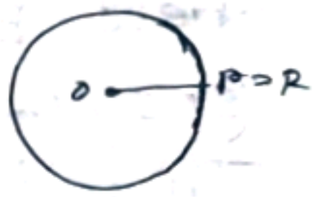


Fig: ii

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Lecture: 05

(iii)

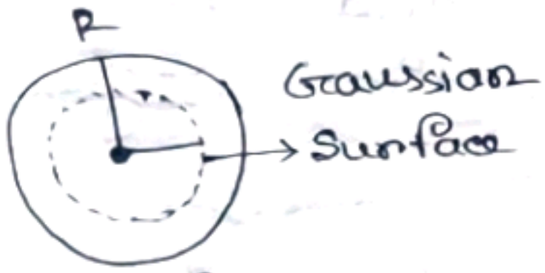


Fig: (iii)

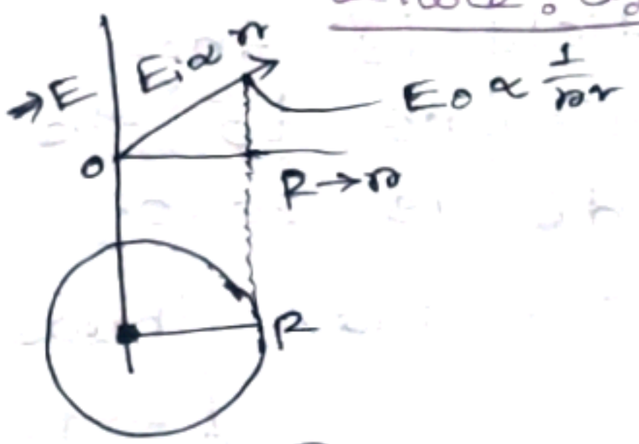


Fig: (iv)

Let p' is a point inside the sphere fig (ii) p' is at a distance r from the centre O. Draw a concentric sphere of radius r ($r < R$) with centre O. This is the gaussian surface.

Total charge enclosed by the gaussian surface,

$$q' = \frac{4}{3}\pi r^3 . \rho$$

$$= \frac{4}{3}\pi r^3 . \frac{q}{4/3\pi R^3}$$

$$= q . \frac{r^3}{R^3}$$

Here, ρ = charge density $= \frac{q}{4/3\pi R^3}$

The outward flux through the surface of the sphere of radius r is

$$\oint \bar{E} . d\bar{s} = E \oint d\bar{s} = E (4\pi r^2) \quad \text{——(i)}$$

Applying gauss's law,

$$\oint \bar{E} . d\bar{s} = \frac{q'}{\epsilon_0} = \frac{q}{\epsilon_0} \frac{r^3}{R^3} \quad \text{——(ii)}$$

Combining (i) and (ii) $\Rightarrow$

$$E(4\pi r^2) = \frac{q}{\epsilon_0} . \frac{r^3}{R^3}$$

$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \times \frac{qr}{R^3}$$

Thus, $E \propto r$ at the centre of the sphere $E = 0$ in fig (iii) shows the variation of electric field $\bar{E}$ as a function of radial distance r.

Question:

An electric dipole when place in an electric field exert a torque of 3.46×10^{-3} Nm and stores in it a potential energy of 2×10^{-3} J. Find out the angle between the electric field and the axis of the dipole?

Soln:

We know,

$$\tau = PE \sin\theta$$

$$\Rightarrow \sin\theta = \frac{\tau}{P \cdot E}$$

$$= \frac{3.46 \times 10^{-3}}{2 \times 10^{-3}}$$

$$= 1.73$$

$$\Rightarrow \theta = \sin^{-1}(1.73)$$

Ans

Here,

$\tau = 3.46 \times 10^{-3}$ Nm

Potential energy,

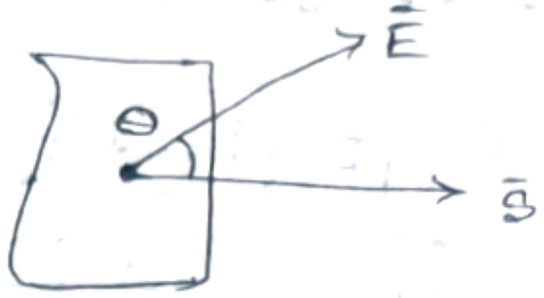
$U = PE = 2 \times 10^{-3}$ J

10) Electric flux:

The electric flux is define as the total number of electric lines of force crossing through the given area. It is denoted by ϕ_e. If the electric field is uniform the electric flux passing through a surface of vector area $\bar{S}$ is

$$\phi_E = \bar{E} \cdot \bar{S} = ES\cos\theta$$

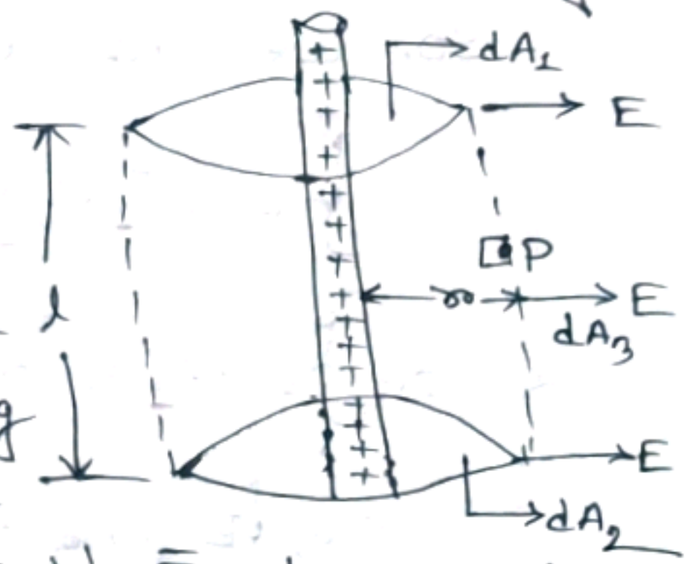
Where E is the electric field S is the area of surface and θ is the angle between the Electric field lines and the normal to S.



Date : 30 · 11 · 21 (Tues) Class : 06

Find an expression for magnitude of the electric field E at distance r from the axis of a long charged plastic rod with linear charge density λ.

Soln:



Let us consider a long plastic rod is uniformly charged having charge per unit length λ. We have find out an electric field $\vec{E}$ at a point 'p' at a distance r from the axis of the rod imagine a gaussian cylindrical surface of arbitary length l of the rod over the point p.

Now,

charge per unit length = λ

length of the cylinder = l

Thus, Total charge enclosed by the cylinder, $q = l\lambda$

Now, According to Gauss's law,

$$\epsilon_0 \oint \bar{E}\, d\bar{l} = q$$

$$\Rightarrow \epsilon_0 \int \bar{E}\, d\bar{l} = l\lambda$$

$$\Rightarrow \oint \bar{E}\, d\bar{l} = \frac{l\lambda}{\epsilon_0}$$

Now, total area of the gaussian cylindrical surface is the sum of the all three portion dA_1, dA_2 and dA_3.

So, we can write,

$$\oint \bar{E}\, d\bar{A}_1 + \oint \bar{E}\, d\bar{A}_2 + \oint \bar{E}\, d\bar{A}_3 = \frac{l\lambda}{\epsilon_0}$$

$$\text{or, } \oint E\, dA_1 \cos 90^\circ + \oint E\, dA_2 \cos 90^\circ + \oint E\, dA_3 \cos 0 = \frac{l\lambda}{\epsilon_0}$$

$$\text{or, } \oint E\, dA_3 = \frac{l\lambda}{\epsilon_0}$$

$$\text{or, } E \oint dA_3 = \frac{l\lambda}{\epsilon_0}$$

$$\text{or, } E \times 2\pi r l = \frac{l\lambda}{\epsilon_0} \qquad \left[\because \oint dA_3 = 2\pi r l\right]$$

$$\text{or, } E = \frac{\cancel{2\pi r}}{2\pi r l \epsilon_0}\, l\lambda$$

$$\therefore E = \frac{\lambda}{2\pi r \epsilon_0} \qquad \text{---------- (1)}$$

If is the expression for electric field for long charge plastic rod or long bar from eq^{n} ①, we can write,

$$E \propto \frac{1}{r}$$

∴ Electric field is inversely proportional to the distance.

☞ Electric field due to a uniform charged plate:

Consider a uniformly charged plane of infinite extent having charge σ per unit area of the surface. An area ΔA of the plane can be enclosed by a gaussian cylinder of the same cross-section fig ①.

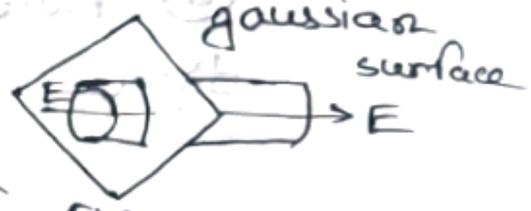


fig: 01

Now, E is uniform and parallel to $d\bar{a}$ on the ends and perpendicular to $d\bar{a}$ on the curved surface, hence we obtain from gauss's law,

$$\oint \bar{E}\, d\bar{a} = \frac{1}{\epsilon_0} \cdot q$$

$$\Rightarrow \oint \bar{E}\, d\bar{a} = \frac{1}{\epsilon_0} \int_S \sigma\, da$$

$$\Rightarrow E\Delta A + E\Delta A = \frac{1}{\epsilon_0} \cdot \sigma \Delta A \qquad \left[\because \oint da = \Delta A\right]$$

$$\Rightarrow 2E\Delta A = \frac{\sigma \Delta A}{\epsilon_0}$$

$$\Rightarrow E = \frac{\sigma}{2\epsilon_0}$$

$$\therefore E = \frac{\sigma}{2\epsilon_0}$$

Date : 02.12.21

Lecture : 07

Exam :

1. State and explain gauss's law.
2. An electric dipole of moment 2×10^{-8} Cm is placed in a uniform field of intensity 1.5×10^{5} NC^{-1}.

(i) What maximum torque does the field exert on the dipole

(ii) How much work is done on turning the dipole end to end ?

Answer :

02 no

(i) We know,

Torque, $\tau = PE \sin\theta$

$\tau_{(max)} = PE$ [$\because \theta = 90^\circ$]

$= 2 \times 10^{-8} \times 1.5 \times 10^{5}$

$= 3 \times 10^{-3}$ Nm

Ans

Here,

$P = 2 \times 10^{-8}$ Cm

$E = 1.5 \times 10^{5}$ NC^{-1}

(11) Since, The dipole dipole is turning on the one end to another end, so, the $\theta = 180°$.

Now, we know,

Work, $U = PE\cos\theta$

$= 2\times10^{-8}\times1.5\times10^{5}\times\cos180°$

$\therefore U = -3\times10^{-3}$ Nm

Ans.

Here,

$P = 2\times10^{-8}$ cm

$E = 1.5\times10^{5}$ NC^{-1}

→ 2nd ←

Date: 09.12.22

Lecture: 08

Capacitor: A capacitor is an electric device consisting of two conductors separated by an insulating or di-electric medium and carrying equal and opposite charges.

Capacitance: If a charge q is given to an isolated conductor its voltage is increased by an amount V. The charge q and potential difference V for a capacitor is proportional to each other

$\therefore q \propto V$

$\Rightarrow q = cV$.

The proportionally constant C is called the capacitance of the capacitor. The unit of capacitance is Farad(F). $1F = 10^6 \mu F$.

$1 \mu F = 10^{-6} F$

$1 pf = 10^{-12} F$

Di-electric: A di-electric is an insulator materials in which all the electrons are tightly bound to the nucleus of the atom. There are no free electron to carry current. Ebonite, Mica and oil and few examples of dielectrics, the electrons are not ~~free to~~ move under the influence of an external field.

Di-electric constant: If C is the capacitance of the capacitor when a di-electric materials is present and C_0 is the capacitance when no di-electric materials is present then the ratio $\frac{C}{C_0}$ is called the di-electric constant K of dielectric materials.

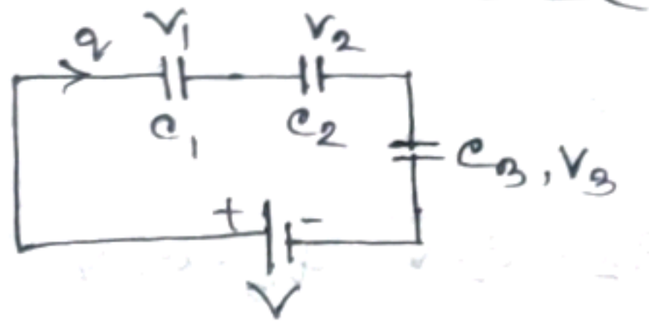
$\therefore K = \frac{C}{C_0}$

☞ There are two types of capacitors.

(i) Capacitors in series.

(ii) Capacitors in Parallel.

☞ Capacitor in series :



Here, $V = V_1 + V_2 + V_3$

and, $V_1 = \frac{q}{C_1}$

$V_2 = \frac{q}{C_2}$

$V_3 = \frac{q}{C_3}$

$\therefore V = \frac{q}{C_1} + \frac{q}{C_2} + \frac{q}{C_3}$ ——— (i)

$\because q = VC_s$

$\Rightarrow V = \frac{q}{C_s}$

Again, $V = \frac{q}{C_s}$ ——— (ii)

From (i) and (ii) $\Rightarrow$

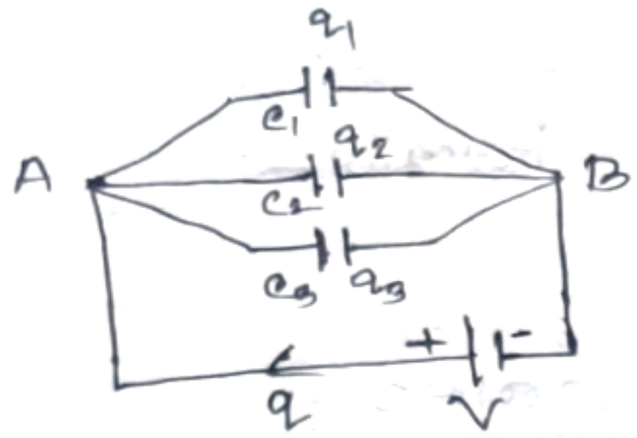
$$\frac{q}{C_s} = \frac{q}{C_1} + \frac{q}{C_2} + \frac{q}{C_3}$$

$$\Rightarrow \frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\therefore \boxed{\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}}$$

Ans.-

Capacitors in Parallel:



Here, $q = q_1 + q_2 + q_3$

and, $q_1 = C_1 V$, $q_2 = C_2 V$, $q_3 = C_3 V$

$\therefore q = C_1 V + C_2 V + C_3 V$ ——— (I)

Again, $q = C_p V$ ——— (II)

Comparing (I) and (II) $\Rightarrow$

$C_p V = C_1 V + C_2 V + C_3 V$

$\Rightarrow \boxed{C_p = C_1 + C_2 + C_3}$

Question:

Two capacitor have a capacity of 5 MF, when connected in parallel and 1.2 MF when connected in series. Calculate the individual capacitance.

Soln: Let, two capacitor are C_1 and C_2.

When, C_1 and C_2 are in parallel, then

$C_1 + C_2 = 5$

$\Rightarrow C_1 = 5 - C_2$ ——— (i)

Again

When C_1 and C_2 are connected in series,

Then,

$$\frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{1.2}$$

$$\Rightarrow \frac{C_1 + C_2}{C_1 C_2} = \frac{1}{1.2}$$

$$\Rightarrow \frac{5}{C_1 C_2} = \frac{1}{1.2}$$

$$\Rightarrow C_1 C_2 = 6$$

$$\Rightarrow (5 - C_2) \cdot C_2 = 6$$

$$\Rightarrow 5C_2 - C_2^2 = 6$$

$$\Rightarrow C_2^2 - 5C_2 + 6 = 0$$

$$\Rightarrow C_2^2 - 3C_2 - 2C_2 + 6 = 0$$

$$\Rightarrow C_2(C_2 - 3) - 2(C_2 - 3) = 0$$

$$\Rightarrow (C_2 - 3)(C_2 - 2) = 0$$

$$\Rightarrow C_2 = 2, 3$$

Now, putting the value of C_2 in equ^2 —(1), we get,

$C_1 = 5 - C_2$ [when $C_2 = 2$]

$= 5 - 2$

$= 3$

$\therefore C_1$ and $C_2 = (3,2)$ and $(3,2)$ Ans

$C_1 = 5 - 3$ [when, $C_2 = 3$]

$= 2$

$\therefore C_1 = 2$

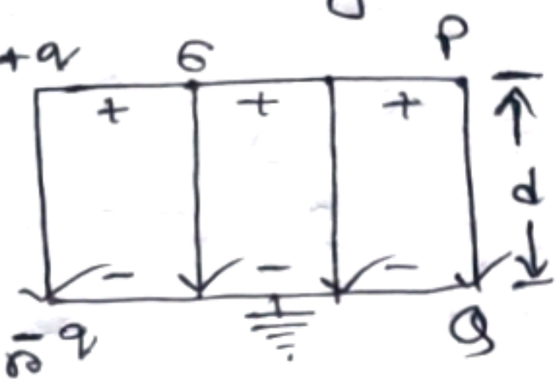
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☞ Capacitance of a parallel plate capacitor.

Or, Prove that $c = \frac{\epsilon \cdot A}{d}$, where symbols are their usual meaning.

Answer:

Let us consider a parallel plate capacitor each plate of area A and separated by a distanced. The medium between the plates is air, a charge $+q$ is given to the plate P it induces a charge $-q$ on the upper plate of the earthed plate $+q$. d is kept small compared with the plate dimension.



By the application of the gauss's law, the electric field at point between the plate is $E = \frac{\sigma}{\epsilon_0}$.

$\because E = \frac{q}{\epsilon_0}$

where, σ = charge density ~~of~~ q/A

$\therefore \sigma = \frac{q}{A}$

$\Rightarrow q = \sigma A$

Potential difference between the plates P and Q is,

$$V = -\int_d^0 E \cdot dr$$

$$= -\int_d^a \frac{\sigma}{\epsilon_0} \cdot dr$$

$$= -\frac{\sigma}{\epsilon_0} [r]_d^0$$

$$\boxed{V = \frac{\sigma d}{\epsilon_0}}$$

∴ The capacitance of a parallel plate capacitor is, $C = \frac{q}{V}$

$$C = q \cdot \sigma A \times \frac{\epsilon_0}{\sigma d}$$

$$\boxed{\therefore C = \frac{A\epsilon_0}{d}}$$

$\because q = \sigma A$

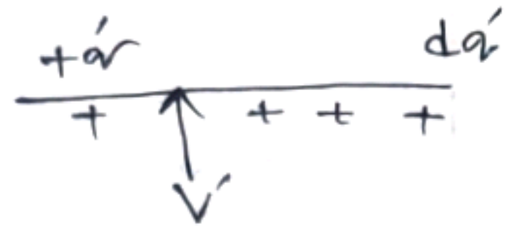
$V = \frac{\sigma d}{\epsilon_0}$

If we put a medium of di-electric constant K, in the free surface between the two plates, then we get,

$$\boxed{C = \frac{K\epsilon_0 A}{d}}$$

☞ Energy stored in a charged capacitor.

Or, Proove that, $u = \frac{1}{2} \epsilon_0 E^2$, where symbols are their usual meaning.



Answer:

Let q' be the charge and v' the potential difference established between the plates of the capacitor at any instant during the process of charging. If an additional charge dq' is given to the plates, the work done by the battery is given by

$$dw = v' dq'$$

$$\because V' = \frac{q'}{C}$$

$$= \frac{q'}{C} dq'$$

Total work done to charge a capacitor to a charge q is

$$w = \int_0^q dw = \int_0^q \frac{q'}{C} dq'$$

$$= \frac{1}{C} \int_0^q q' \, dq'$$

$$\therefore w = \frac{1}{2} \frac{q^2}{C}$$

This work done is stored as electrostatic potential energy in the capacitor. is

$$U = \frac{1}{2}\frac{q^2}{C}$$

$$= \frac{1}{2}\frac{C^2V^2}{C}$$

$$\therefore U = \frac{1}{2}CV^2$$

Consider a parallel plate capacitor of area A and plate separation d, The energy of capacitor,

$$U = \frac{1}{2}CV^2 = \frac{1}{2}\left(\frac{\epsilon_0 A}{d}\right)\cdot V^2$$

Volume of the space between the plate = Ad.
Energy density u is the potential energy per unit volume,

$$u = \frac{U}{Ad}$$

$$= \frac{\frac{1}{2}\left(\frac{\epsilon_0 A}{d}\right)V^2}{Ad}$$

$$= \frac{\epsilon_0 A V^2}{2d} \times \frac{1}{Ad}$$

$$\therefore u = \frac{1}{2}\epsilon_0\left(\frac{V}{d}\right)^2$$

$$\because E = \frac{V}{d}$$

OR, $$u = \frac{1}{2}\epsilon_0 E^2$$

Where, u = Energy density.

Di-electric and gauss's law:

Let us consider two parallel plate capacitors, one of them with dielectric and another without dielectric fig(a) shows an usual parallel plate capacitor without dielectric.

let charge be Q on the plate, with proper sign and A be the area of each plate, E_0 is the electric field without di-electric.

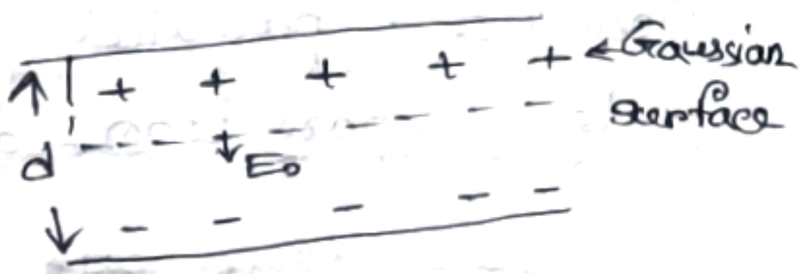


fig:(a)

Therefore, the gaussis law gives,

$$\epsilon_0 \oint \bar{E} \cdot d\bar{s} = \epsilon_0 E_0 A = Q$$

$$\text{or, } E_0 = \frac{Q}{\epsilon_0 A} \quad \text{---(i)}$$

Now, place the di-electric as shown in fig (b) then the net charge enclosed by gaussian surface is $Q - Q'$, therefore with dielectric the gauss's law as,

$$\epsilon_0 \oint \bar{E} \cdot d\bar{s} = Q - Q' \quad \text{---(ii)}$$

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before: 10 $\Rightarrow \epsilon_0 E A = Q - Q'$

$$\Rightarrow E = \frac{Q - Q'}{\epsilon_0 A}$$

$$\therefore E = \frac{Q}{\epsilon_0 A} - \frac{Q'}{\epsilon_0 A} \quad \text{---(iii)}$$

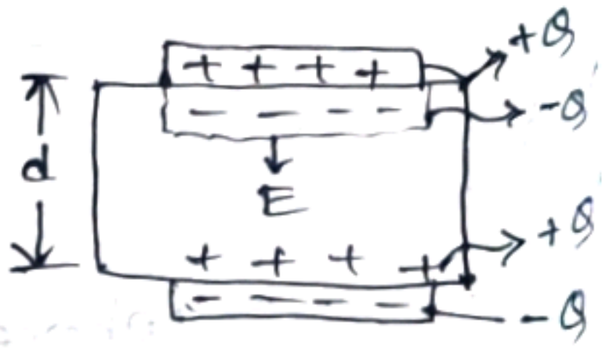


Fig:(b)

Son which $-Q$ is the induced charge due to polarization but we know dielectric constant

$$K = \frac{E_0}{E}$$

$$\Rightarrow E = \frac{E_0}{K} \longrightarrow (iv)$$

From equ^n (i) and (iv), we get.

$$E = \frac{E_0}{K}$$

$$= \frac{\frac{Q}{\epsilon_0 A}}{K} \quad \left[\because E_0 = \frac{Q}{\epsilon_0 A}\right]$$

$$\therefore E = \frac{Q}{K\epsilon_0 A} \longrightarrow (v)$$

Subtitution this value of E in equ^n (iii), we get,

$$\frac{Q}{K\epsilon_0 A} = \frac{Q}{\epsilon_0 A} - \frac{Q'}{\epsilon_0 A} \longrightarrow (vi)$$

$$\therefore Q - Q' = \frac{Q}{K} \longrightarrow (vii)$$

From equ^n (ii) and (vi), we get,

$$\epsilon_0 \oint \bar{E} \cdot d\bar{s} = \frac{Q}{K}$$

$$\therefore \epsilon_0 \oint K \bar{E} \cdot d\bar{s} = Q$$

This is the gauss's Law with di-electric.

Question:

show that, $\bar{D} = \epsilon_0 \bar{E} + \bar{P}$

Or, Find out the relation between electric displacement and electric polarization:

Soln:

The polarization P is defined as the induced surface charge per unit area.
i.e., surface density of bound charges in di-electric.

$\therefore$ polarization, $P = \frac{q}{A}$ —①

The di-electric constant k for any medium is $K = \frac{E_0}{E}$

$\Rightarrow E = \frac{E_0}{K}$

$= \frac{\frac{q}{\epsilon_0 A}}{K}$ $\left[\because E_0 = \frac{q}{\epsilon_0 A}\right]$

$\therefore E = \frac{q}{K\epsilon_0 A}$ —— (ii)

Again, $E = \frac{q}{\epsilon_0 A} - \frac{q'}{\epsilon_0 A}$ —— (iii)

From eqn (II) and (III), we get,

$$\frac{q}{K\epsilon_0 A} = \frac{q}{\epsilon_0 A} - \frac{q'}{\epsilon_0 A}$$

$$\Rightarrow \frac{q}{\epsilon_0 A} = \frac{q}{K\epsilon_0 A} + \frac{q'}{\epsilon_0 A}$$

$$\Rightarrow \frac{q}{A} = \epsilon_0\left(\frac{q}{K\epsilon_0 A}\right) + \frac{q'}{A} \quad \text{---(IV)}$$

From, (I), (II) and (IV), we get,

$$q/A = \epsilon_0 E + P \quad \text{---(V)}$$

But, electric displacement

$$D = \frac{q}{A}$$

$$\therefore D = \epsilon_0 E + P \quad \text{---(VI)}$$

As E and P on the right hand side of eqn (VI) are vectors, D is also a vector. Hence,

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

(Hence proved)

Question : 01

The area of each plate of an air-filled parallel plate capacitor is $1.1 \times 10^8\ m^2$. What must be the separation between the plates, if the capacitance is to be 1.0 Farad.

Solution:

We know,

$$C = \frac{\epsilon_0 A}{d}$$

$$\Rightarrow d = \frac{\epsilon_0 A}{C}$$

Here,

$\epsilon_0 = 8.85 \times 10^{-12}\ F\,m^{-1}$

$A = 1.1 \times 10^8\ m^2$

$d = ?$

$C = 1\ F$

$$= \frac{8.85 \times 10^{-12} \times 1.1 \times 10^8}{1} \left(\frac{F\,m^{-1} \cdot m^2}{F}\right)$$

$\therefore d = 9.735 \times 10^{-4}\ m$ Ans

Question: 02

A plane parallel plate capacitor has circular plates of radius $r = 10.0\ cm$ separated by distance $d = 1.0\ mm$. How much charge is stored on each plate when their electric potential difference has the value $V = 100\ V$.

Soln:

We know,

$q = cv$ where c is the capacitance and is given by

$$c = \frac{\epsilon_0 A}{d}$$

Here,

$\epsilon_0 = 8.85 \times 10^{-12}\ F m^{-1}$

$A = \pi r^2 = 3.14 \times (0.1\,m)^2$

$= 3.14 \times 10^{-2}\ m^2$

$d = 1\ mm$

$= 10^{-3}\ m$

$$= \frac{8.85 \times 10^{-12}\ F m^{-1} \times 3.14 \times 10^{-2}\ m^2}{10^{-3}\ m}$$

$\therefore c = 2.8 \times 10^{-10}\ F$

Again,

$q = cV$

$= 2.8 \times 10^{-10} \times 100$

$\because V = 100\ V.$

$\therefore$ charge, $q = 2.8 \times 10^{-8}\ C$.

Ans.

Q. A parallel plate capacitor of plate area 115 cm^2 and plate separation 1.24 cm is charged to 85.5 volts. Calculate the free charge on the plates of the capacitor.

Given, $A = 115\ cm^2 = 115 \times 10^{-4}\ m^2$

$d = 1.24\ cm = 1.24 \times 10^{-2}\ m$

$\epsilon_0 = 8.85 \times 10^{-12}\ F m^{-1}$

$$\therefore c = \frac{\epsilon_0 A}{d}$$

$$= \frac{8.85 \times 10^{-12} \times 115 \times 10^{-4}\ m^2}{1.24 \times 10^{-2}\ m}$$

$= 8.21 \times 10^{-12}\ F$

Now, the free charge,

$q = cV$

$= 8.21 \times 10^{-12} \times 85.5\ V$

$= 7.02 \times 10^{-10}\ C$

Ans

Date : 21.12.21

Lecture: 11

Question:

A 3.55 μF capacitor C_1 is charged to a potential difference $V_0 = 6.30$ V using a battery. The charging battery is then removed and the capacitor is connected to uncharged 8.95 μF capacitor C_2 in series as in the figure. After the switch S is closed charge flows from C_1 to C_2 until an equilibrium is established with both capacitor at the same potential difference.

(i) What is the common potential difference V?

(ii) What is the energy stored in the electric field before and after the switch S is closed?

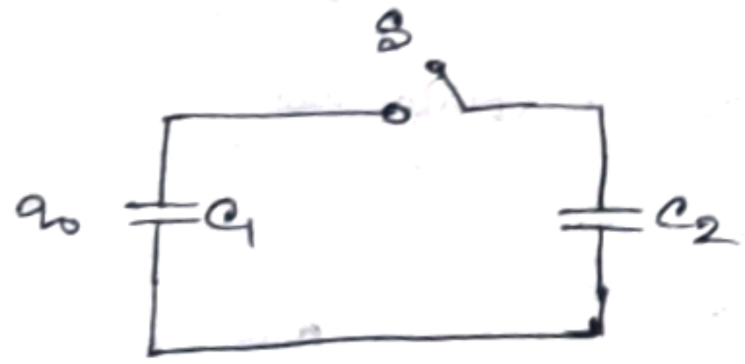


Fig: 1

Solⁿ:

(i) The original charge q_0 is now shared by two capacitors.

So, $q_0 = q_1 + q_2$

Applying the relation $q = cv$, to each term of this equation fields,

$$c_1 V_0 = c_1 v + c_2 v$$

$$= v(c_1 + c_2)$$

$$\Rightarrow V = \frac{c_1 V_0}{c_1 + c_2}$$

$$= \frac{3.55\,\mu F \times 6.30\,V}{(3.55 + 8.95)\,\mu F}$$

$$V = 1.79\,V \quad \text{Ans.}$$

Here,
$c_1 = 3.55\,\mu F$
$c_2 = 8.95\,\mu F$
$V_0 = 6.30\,V$

(ii) The Initial stored energy:

$c_1 = 3.55\,\mu F = 3.55 \times 10^{-6}\,F$

$U_i = \frac{1}{2} c_1 V_0^2 = \frac{1}{2} \times 3.55 \times 10^{-6}\,F \times (6.3\,V)^2$

$= 70.45 \times 10^{-6}\,J$

And, the final energy,

$c_1 + c_2 = (3.55 + 8.95)\,\mu F = 12.5 \times 10^{-6}\,F$

$U_f = \frac{1}{2} c_1 v^2 + \frac{1}{2} c_2 v^2$

$= \frac{1}{2} v^2 (c_1 + c_2) = \frac{1}{2} \times (1.79)^2 \times 12.5 \times 10^{-6}\,F$

$\therefore U_f = 2.0 \times 10^{-5}\,J$ Ans.

Question:

A parallel plate capacitor is filled with two dielectrics of same direction but different dielectric constant K_1 and K_2 respectively as shown fig-(a) and fig-(b). Calculate its capacitance.

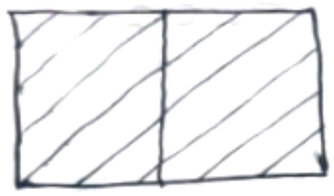


Fig-(a)



fig-(b)

Soln:

From fig (a), the arrangement is simply two capacitors in parallel and each capacitor will have the area $A/2$.

$\therefore$ Total capacitance,

$$C = C_1 + C_2$$

$$= \frac{\epsilon_0 A/2 \cdot K_1}{d} + \frac{\epsilon_0 A/2 \cdot K_2}{d}$$

$$= \frac{\epsilon_0 A}{d}\left(\frac{K_1}{2} + \frac{K_2}{2}\right)$$

$$\therefore C = \frac{\epsilon_0 A}{d}\left(\frac{K_1 + K_2}{2}\right).$$

The arrangement of fig (b) can be regarded as two capacitors in series.

Each capacitor will be of area A and separated $d/2$.

Therefore, the total capacitance,

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$= \frac{d}{2}\left(\frac{1}{\epsilon_0 A K_1} + \frac{1}{\epsilon_0 A K_2}\right)$$

$$= \frac{d}{2\epsilon_0 A}\left(\frac{1}{K_1} + \frac{1}{K_2}\right)$$

$$= \frac{d}{2\epsilon_0 A}\left(\frac{K_1 + K_2}{K_1 K_2}\right)$$

$$= \frac{2\epsilon_0 A}{d}\left(\frac{K_1 K_2}{K_1 + K_2}\right).$$

Question:

The plates of parallel plate capacitor have an area of 90 cm^2 each and separated by 2.5mm, the capacitor is charged by connecting it to a 400V supply. How much electrostatic energy is stored by the capacitors?

Sol^n:

Given that,

$$A = 90\ cm^2$$

$$= 90 \times 10^{-4}\ m^2$$

$$d = 2.5\ mm$$

$$= 2.5 \times 10^{-3}\ m$$

$$\epsilon_0 = 8.85 \times 10^{-12}\ F\,m^{-1}$$

$$V = 400\ V$$

We know,

$$C = \frac{\epsilon_0 A}{d}$$

$$= \frac{8.85 \times 10^{-12}\ F\,m^{-1} \times 90 \times 10^{-4}\ m^2}{2.5 \times 10^{-3}\ m}$$

$\therefore$ capacitance, $C = 3.186 \times 10^{-11}\ F$

Again,

Electrostatic energy, $U = \frac{1}{2} C V^2$

$$= \frac{1}{2} \times 3.19 \times 10^{-11}\ F \times (400\ V)^2$$

$\therefore$ Electrostatic energy, $U = 2.55 \times 10^{-6}\ J$.

Ans.

Date: 29.012.21 Lecture: 12

A parallel - plate capacitor of plate area $10^{-2}\,m^2$ and separation $10^{-2}\,m$ is charged to 100volts. Then after removing the charging battery a slab of insulating material of thickness $0.5 \times 10^{-2}\,m$ relative permittivity 7 is inserted between the plates. Calculate the free charge on the plates of two capacitor, electric field intensity in air, electric field intensity in the di-electric, potential difference between the plates and the capacitance (with dielectric present).

Soln:

The capacitance C_0 before the slab is inserted is,

$$C_0 = \frac{\epsilon_0 A}{d}$$

$$= \frac{8.85\times10^{-12}\ F\,m^{-1} \times 10^{-2}\,m^2}{10^{-2}\,m}$$

$$\therefore C_0 = 8.9\times10^{-12}\ F.$$

Here,
$\epsilon_0 = 8.85\times10^{-12}\ F\,m^{-1}$
$A = 10^{-2}\,m^2$
$d = 10^{-2}\,m$

Therefore, the free charge, $q = C_0 V_0$

$$= 8.9\times10^{-12}\,F \times 100\,V$$

$$\therefore q = 8.9\times10^{-10}\,C$$

$V_0 = 100\,V$

Electric field intensity,

$$E_0 = \frac{V_0}{d}$$

$$= \frac{100\ V}{10^{-2}\ m}$$

$$\therefore E_0 = 1 \times 10^4\ Vm^{-1}$$

Here,
$V_0 = 100\ V$
$d = 10^{-2}\ m$

Again,

Dielectric field intensity in Dielectric is

$$E = \frac{E_0}{\varepsilon_0}$$

Here, $\varepsilon_0 = 7$

$$= \frac{1 \times 10^4\ Vm^{-1}}{7}$$

$$\therefore E = 1.43 \times 10^3\ Vm^{-1}$$

∴ The potential difference between the plates with dielectric present is,

$$V = E_0(d-t) + Et$$

$$= 10^4\left(10^{-2} - 0.5 \times 10^{-2}\right) + 1.43 \times 10^3 \times 0.5 \times 10^{-2}$$

$$= 50 + 7.15$$

$$= 57.15\ V$$

$$\therefore V = 57\ V$$

∴ The capacitance (with di-electric present),

$$C = \frac{q}{V}$$

$$= \frac{8.9 \times 10^{10}\, e}{57\ V}$$

$$= 16 \times 10^{-12}\ F$$

$$\therefore C = 16PF \quad .$$

Ans.

Polarization:

Let, the small displacement of the charge q_i in an induced dipole be $\bar{l}_i$, then the electric moment of dipole is $\bar{P}_i = q_i \bar{l}_i$.

Now, there are n dipoles in a small volume ΔV around a given point, then the polarization at that point is defined as the vector sum of the n dipoles divided by the volume ΔV.

In other words,

$$\bar{P} = \frac{1}{\Delta V} \sum_{i=1}^{n} \bar{P}_i$$

$$= N \bar{p}$$

Where, N is the number of dipole per unit volume. and, $\bar{p}$ is the average dipole moment.

Question: What are Electronic susceptibility and permittivity ?

Answer:

Electronic susceptibility: The polarization $\bar{P}$ in a homogeneous isotropic dielectric is dependent on the nature of the dielectric, it has the same direction as the resultant electric field and also is dependent on the field. These results are summarized by the equation:

$$\bar{P} \propto \bar{E}$$

$$\Rightarrow \bar{P} = \chi_e \bar{E}$$

where, χ_e is a scalar quantity is called the electronic susceptibility of the material. As we know that electric displacement $\bar{D}$ for a dielectric medium is,

$$\bar{D} = \epsilon_0 \bar{E} + \bar{P}$$

$$= \epsilon_0 \bar{E} + \chi_e \bar{E}$$

$$= (\epsilon_0 + \chi_e)\bar{E}$$

$$\therefore \bar{D} = \epsilon \bar{E}$$

where $\epsilon = \epsilon_0 + \chi_e$ and is called the permittivity of material. For air or vacuum, $p = 0$ and so $\chi_e = 0$ and $\epsilon = \epsilon_0$.

Electric Current

(3)

- Electron theory of conductivity ;
- conductors
  - Semiconductors and insulators
  - Superconductors
  - Current and current density
  - Kirchhoff's Law and its applications.

Date: 26.12.21 Lecture: 13

Question: What is electric current and density?

Electric current: If a charge of pass through any cross-section of conductor in time t, then the curre is defined as

$$i = \frac{q}{t}$$

where q is in coulomb and t is in second, then is in ampere.

$$\therefore i = \frac{dq}{dt}$$

Current density: The current density of any point is defined as the quantity of charge passing per second through a unit area taken perpendicular to the direction of the flow of the charge at that point. It is a vector quantity. If the curre is flowing uniformly across a conductor of cross sectional area A. The magnitude of the curren density is given by

$$j = \frac{q/t}{A} = \frac{i}{A}$$

Unit of current density is Am^{-2}.

molecules of a perfect gas in a container.

2. The free electrons make collisions from time to time with fixed positive ions in the lattice and also among themselves. We may neglect the collisions between electrons in comparison with the collisions between electrons and ion cores.

3. In the absence of electric field, the random motion of free electrons is equally probable in all directions so that current density vector is zero.

4. When an external electric field is applied, the electrons drift slowly with some average velocity, known as *average drift velocity*, in the direction opposite to that of electric field. The drift velocity of free electrons is superimposed over their random velocity. In Fig. 6.3, the continuous solid lines show a possible random path followed by an electron in the absence of an applied field. *The dashed lines show the electron path in the presence of the electric field* **E**. Note the steady drift in the direction of $-\mathbf{E}$. The drift velocity is much smaller in magnitude than average random velocity of free electrons.

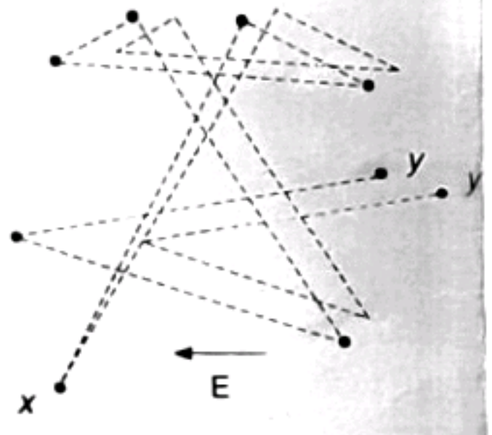


Fig. 6.3

5. The average distance traversed by a free electron between two successive collisions with the positive ions is called *mean free path* and is denoted by λ.

If an electron of mass m and charge e is placed in an electric field E, it experiences an acceleration a given by

$$a = eE/m.$$

Let τ be the average time-interval between two successive collisions of an electron with positive ions.

Increase in drift velocity in time $\tau = a\tau = eE\tau/m$

This drift velocity is reduced to zero at the next collision and again builds up to the value $eE\tau/m$ before suffering another collision and so on.

$$\therefore \quad \left.\begin{matrix}\text{the average drift velocity} \\ \text{through the conductor}\end{matrix}\right\} = v_d = \frac{eE\tau}{2m}$$

Let n be the number of conduction electrons per unit volume.

$$\text{Current density} = J = nev_d = \frac{ne^2E\tau}{2m}$$

But $J = \sigma E$ where σ is the electrical conductivity of the metal

$$\therefore \quad \sigma = \frac{J}{E} = \frac{ne^2\tau}{2m}$$

This is the expression for the electrical conductivity of a metal.

(Note. Some authors write the above equation without the factor 1/2.)

Resistivity $= \rho = 1/\sigma = 2m/ne^2\tau$

Let λ be the mean free path and v the r.m.s. thermal velocity of electrons. Then $\tau = \lambda/v$

$$\text{Electrical conductivity, } \sigma = \frac{ne^2\lambda}{2mv}$$

The kinetic energy of a free electron, $\frac{1}{2}mv^2 = \frac{3}{2}kT$.

$$\therefore \quad \sigma = \frac{ne^2\lambda v}{2mv^2} = \frac{ne^2\lambda v}{6kT}$$

Deduction of Ohm's law From Drude-Lorentz theory

From Drude Lorentz theory, $J = \sigma E = \frac{ne^2\lambda}{2mv}\mathbf{E}$

At a particular temperature, λ and v are constant.

$$\therefore \quad \mathbf{J} \propto \mathbf{E}$$

Let i be the current flowing through the conductor of length l and area of cross-section A and V be the potential difference across it. Then

$$J = \frac{i}{A} \text{ and } E = \frac{V}{l}$$

The above relation becomes

$$\frac{i}{A} \propto \frac{V}{l}$$

For a given conductor, l and A are constant at a given temperature.

$$\therefore \quad i \propto V$$

i.e., at a given temperature, the current flowing through a conductor is proportional to the potential difference between its two ends, which is Ohm's law.

Variation of Electrical Conductivity with Temperature

The electrical conductivity of metals is given by

$$\sigma = \frac{ne^2\tau}{2m}$$

According to the kinetic theory of gases (here we are dealing with electron-gas) the time-interval τ between two collisions is inversely proportional to the square-root of the absolute temperature :

Question: what is drift velocity? Relation between current density and drift velocity?

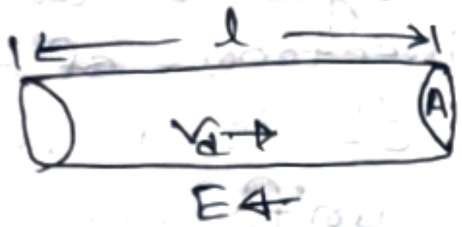
Or, prove that, $v_d = J/ne$, where the symbols are their usual meaning.

Or, Find an equation for drift velocity when an electric field is applied on the conductor.

Answer:

Drift velocity: A potential difference between the two ends of the conductor is established the free electrons drift towards $-E$, that is opposite to the electric field inside the conductor with an average speed. This average speed is known as drift velocity.
It is denoted by v_d.

$$\therefore v_d = \frac{J}{ne}$$



Let us consider a conductor of length l and cross-sectional area A.

The electrons move with average velocity v_d towards $-E$. If n is the number of electron per unit volume of the conductor, then the total number of electron inside volume Al is given by,

$$N = nAl \quad \text{---} (1)$$

Total charge flowing ~~inde~~ inside the conductor

$$q = Ne \quad \text{——— (II)}$$

From (I) and (II) $\Rightarrow$

$$q = n e A l \quad \text{——— (III)}$$

The average time taken by the electrons to travel through the conductor is

$$t = \frac{l}{v_d}$$

We know; $i = \frac{q}{t}$ $\qquad J = \frac{I}{A}$

$$\therefore \quad = \frac{q}{l/v_d}$$

$$= \frac{q v_d}{l}$$

$$= \frac{n e A l v_d}{l}$$

$$\therefore i = n e A v_d$$

Again, we know, $\quad j = \frac{i}{A}$

$$= \frac{n e A v_d}{A}$$

$$\Rightarrow j = n e v_d$$

$\therefore$ Drift velocity, $\boxed{v_d = \frac{j}{ne}}$

H.W: Conductor, Semiconductor, insulator (Defi)

Answer:

Insulator:

Insulator (e.g wood, Glass etc) are those substances which do not allow the passage of electric current through them in terms of energy band, the valance band is full while the conduction band is empthy. Further the energy gap between valence and conduction band is very large ($\cong$ 15eV) as shown in fig ①.

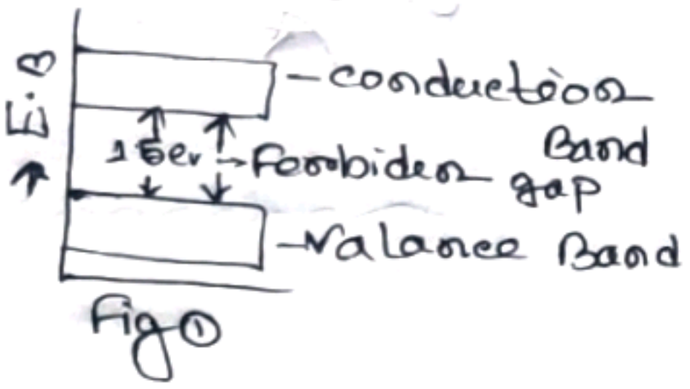


Fig ①

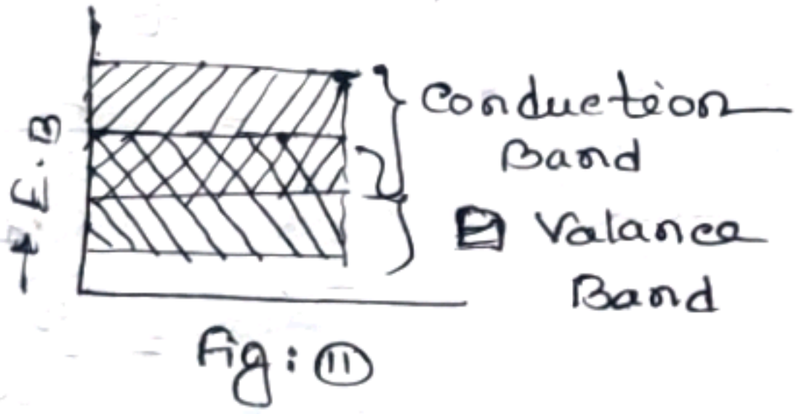


Fig: ⑪

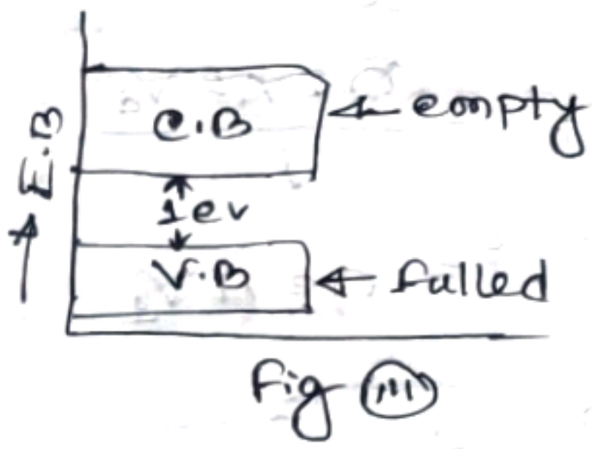


Fig ⑪

Conductor: (Metal):

In conductor e.g. (copper, aluminium) there is no forbiden gap available. The valance and conduction band overlap each other (Fig-2). The electrons from valance band freely enters into the

conduction band, due to the overlapping of the valance and conduction bands a very low potential difference can cause the continuous flow of current.

Semiconductors:

In semiconductors (eg {germanium, silicon, etc) in fig (iii), the forbidden gap is very small. Germanium and silicon are the best examples of semiconductors. The forbidden energy gap is of the order of 0.7 ev for Ge and 1.1 ev for Si. There are no electrons in the conduction band. The valance band is completely filled at 0 k. With a small amount of energy that is supplied the electron can be easily jump from the valance band to the conduction band fig (iii).

Date:

Questions:

Exam: 02.01.22 → 09.01.22 (sunday)

☞ What is electric dipole and its moment?

☞ Find potential energy of a dipole in a uniformly electric field.

☞ State and proves gauss's law of electrostatistic.

☞ Columbs law from it.

☞ Math

Date: 28.12.21

lecture: 19

☞ state and explain kirchhoff's law in a network.

1st law or current law:

In any electrical networks the algebraing sum of current meeting a junction is always zero.

i.e, $\sum i = 0$.

Explain:

let us consider a junction that i_1, i_2, i_3, i_4, i_5, i_6 and i_7 currents are different direction and meeting of a point O. Taking the currents flowing away from the junction

as negative and the currents flowing towards the junction as positive.

we, get,

$$i_1 + i_2 + i_3 + i_4 - i_5 - i_6 - i_7 = 0$$

or, $i_1 + i_2 + i_3 = i_4 + i_5 + i_6 + i_7$

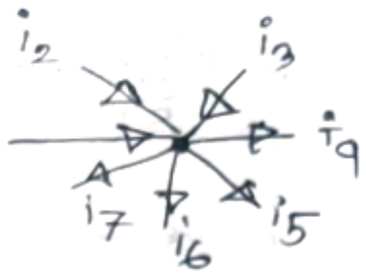


Fig: (a)

i.e, incoming current = outgoing current.

Total current entering any junction of a circuit equal the total current leaving that junction.

Kirchhoff's 2nd law/Voltage law: In a closed circuit, algebraic sum of the products of the current and resistance of the each part of the circuit is equal to the total emf's acting in the circuit.

i.e., $\Sigma iR = \Sigma E$

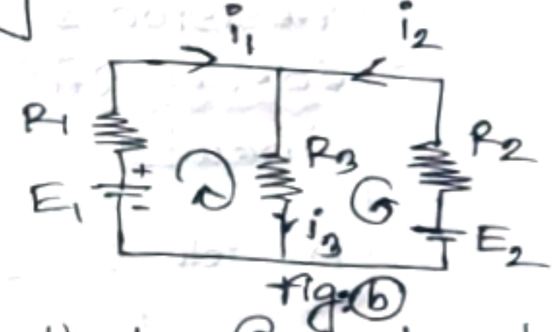


fig: (b)

Explanation:

In the fig (b), we see that for closed circuit formed by E_1, E_2 R_1, R_2, R_3, we have according to second law,

$$E_1 - E_2 = i_1 R_1 - i_2 R_2$$

and also for the closed circuit formed by E_1, R_1, R_3, and E_1.

We have,

$$E_1 = i_1 R_1 + i_3 R_3$$

Similarly, $E_2 = i_2 R_2 + i_3 R_3$

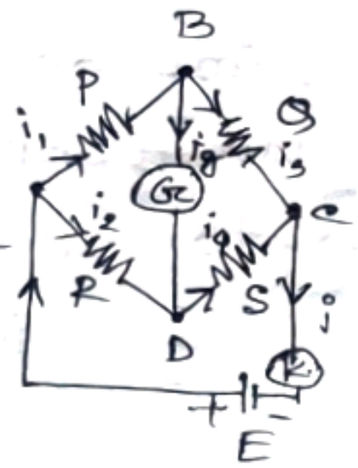
$\therefore$ In general, $\Sigma E = \Sigma iR$

In a closed circuit the 2nd Law is also called Loop theorem.

☞ Find the expression for current in the galvanometer in a whatstones bridge network and hence show that $P/Q = R/S$.

Answer:

Let us consider a wheaston bridge circuit where resistance P, Q, R and S are connected to from a closed circuit.



A cell E and a key K is connected between the points A and C and a galvanometer (G) is connected between the points B and D. The current I goes to the points A and D from the positive ends of the cell and devided into i_1 and i_2 i_1 passes A to B through the resistance P and i_2 passes A to D through the resistance R.

i_g passes B to D through the galvanometer Gr. From the point B, current i_3 passes to C through the resistance Q and from the point D, current i_4 passes to C through the resistance S.

Current i_3 and i_4 joined together in the point C and come back to the cell being current i.

Applying the kirchhoffis 1st law in B and D point, we get,

$$i_1 - i_3 - i_g = 0$$

$$\Rightarrow i_1 = i_3 + i_g \quad \text{——} (1)$$

and $$i_2 + i_g - i_4 = 0$$

$$\Rightarrow i_g + i_2 = i_4 \quad \text{——} (11)$$

Now, we applying the kirchhoff's 2nd law, in closed circuit ABDA and BCDB, we get.

$$i_1 P + i_g G_r - i_2 R = 0 \quad \text{——} (111)$$

and $$i_3 Q - i_4 S - i_g G_r = 0 \quad \text{——} (IV)$$

But, in the balanced position of bridge $i_g = 0$.

So, from eqn (1) and (11), we get,

$$i_1 = i_3$$

and $$i_2 = i_4$$

Now according to eqn (iii) and (iv), we get,

$$i_1 P = i_2 R \quad \text{—(v)}$$

$$i_3 Q = i_4 S \quad \text{—(vi)}$$

By dividing equn (v) and (vi), we get,

$$\frac{i_1 P}{i_3 Q} = \frac{i_2 R}{i_4 S}$$

$$\Rightarrow \frac{P}{Q} = \frac{R}{S} \quad \left[\because i_1 = i_3 \text{ and } i_4 = i_2\right]$$

$$\therefore \boxed{\frac{P}{Q} = \frac{R}{S}}$$

Date: 30.12.21 lecture: 15

Resistance: The property of a substance due to which opposes the flow of electricity through it is called resistance. It is denoted by R. It can be measured by the ohm's law.

According to ohm's law,

$$I = \frac{V}{R}$$

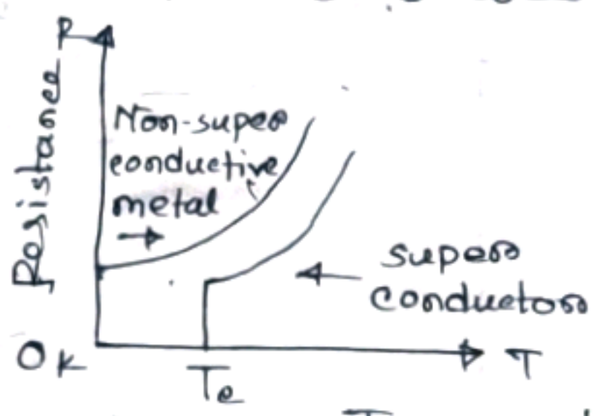
$$\therefore R = \frac{V}{I}$$

where, R resistance, V is potential difference, I is current.

The unit of resistance is ohm (Ω).

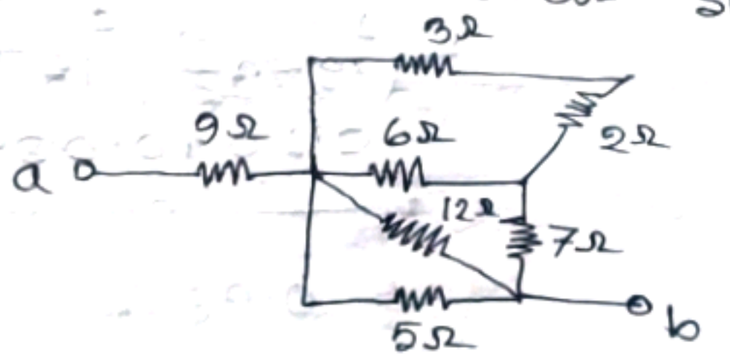
Date: 06.01.22 Lecture: 16

Super conductor: The electrical resistivity of many metals and alloys drops suddenly to zero when the specimen is cooled to a sufficiently low temperature. Often a temperature in the liquid helium range. The phenomena of the reduction in the value of electrical resistivity to zero is known as superconductivity and the meterials which show this propeerty are called the superconductors. Examples of super conductors are Nb_3Ge, Nb_3Al, K_3C_{60} etc.

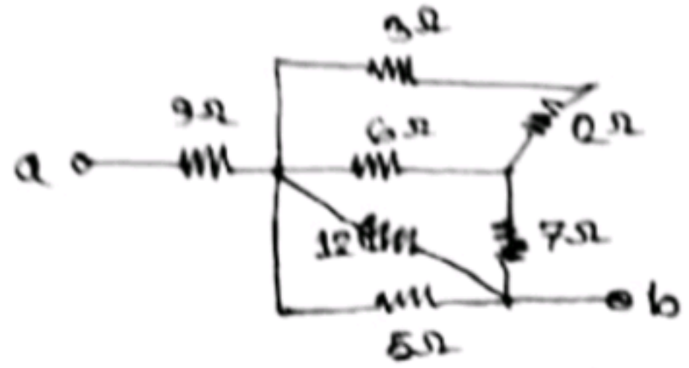


This phenomena was observed first by Heike Kamerlingh Onnes in 1911 0K in mercury. He observed a sudden decrease Temperature in resistivity when pure mercury was cooled down below 4.2K. The resistivity of mercury vanish completely below 4.2 k.

Question: Find the equivalent resistance between points a and b for the combination shown in figure:



Answer:



Here,

3Ω and 2Ω are in series

$$R_s = (3\Omega + 2\Omega) = 5\Omega$$

Again,

R_s and 6Ω are in parallel.

$$\therefore \frac{1}{R_p} = \left(\frac{1}{R_s} + \frac{1}{6}\right)$$

$$= \frac{1}{5} + \frac{1}{6}$$

$$= \frac{11}{30}$$

$$\therefore R_p = 2.73$$

Again, R_p and 7Ω in series.

$$\therefore R_s' = (R_p + 7) = (2.73 + 7) = 9.73\Omega$$

Now,

R_s', 12Ω and 5Ω are in parallel.

Therefore, $\frac{1}{R_p'} = \left(\frac{1}{R_s'} + \frac{1}{12} + \frac{1}{5}\right)$

$$= \left(\frac{1}{9.73} + \frac{1}{12} + \frac{1}{5}\right)$$

$$= 0.10 + 0.08 + 0.2$$

$$= 0.38\Omega$$

$$\therefore R_p' = 2.63\Omega$$

Again, R_p' and 9Ω are in series.

$$\therefore R_T = (R_p' + 9\Omega) = (2.63 + 9)\Omega$$

$$= 11.63\Omega$$ Ans

Question: An electronic circuit is made as shown in fig ①. Calculate the current following through various branches.

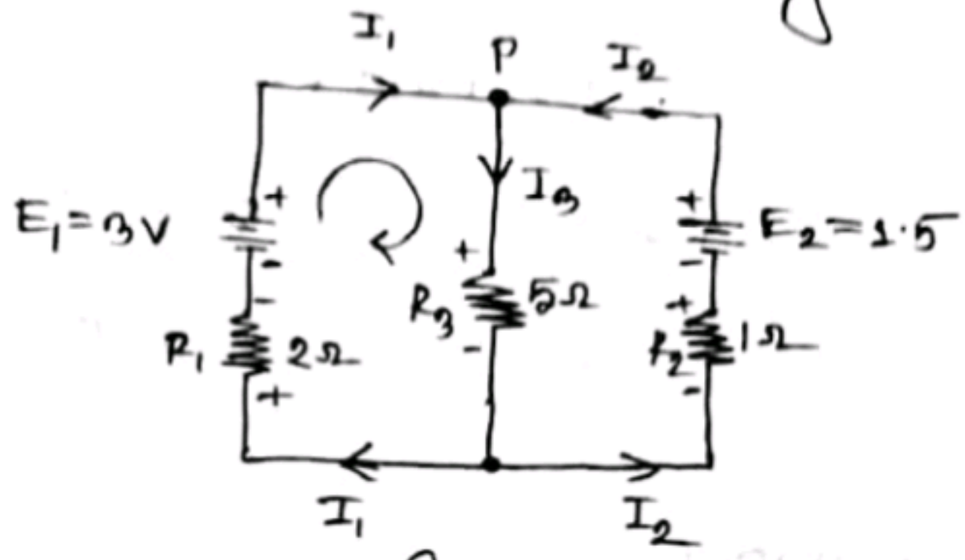


fig-①

Answer: let us consider, the closed circuit, $E_1R_3R_1E_1$ from kirchhoff's 2nd law.

$$E_1 = I_3R_3 + I_1R_1$$

$$\Rightarrow 3 = 5I_3 + 2I_1$$

$$\Rightarrow 2I_1 + 5I_3 - 3 = 0 \quad \text{——} ①$$

For Loop $E_1E_2R_2R_1E_1$, we have,

$$E_1 - E_2 = -I_2R_2 + I_1R_1$$

$$\Rightarrow 3 - 1.5 = I_2 + 2I_1$$

$$\Rightarrow 2I_1 + I_2 - 1.5 = 0 \quad \text{——} ⑪$$

Again, From node P, $I_1 + I_2 = I_3$ —— ⑪

Fooom eqⁿ ①, we get,

$$2I_1 + 5I_3 - 3 = 0$$

$$\Rightarrow 2I_1 + 5(I_1 + I_2) - 3 = 0 \quad [\because I_1 + I_2 = I_3]$$

$$\Rightarrow 2I_1 + 5I_1 + 5I_2 - 3 = 0$$

$$\Rightarrow 7I_1 + 5I_2 - 3 = 0 \quad \text{——— (iv)}$$

Therefore,

$$7I_1 + 5I_2 - 3 = 0 \quad \text{——— (iv)}$$

$$2I_1 + I_2 - 1.5 = 0 \quad \text{——— (ii)}$$

Now,

$$\frac{I_1}{-7.5+3} = \frac{I_2}{-10.5+6} = \frac{1}{7-10}$$

$$\Rightarrow \frac{I_1}{-4.5} = \frac{I_2}{-9.5} = \frac{1}{-3}$$

Now,

$$\frac{I_1}{-7.5-3} = \frac{I_2}{-6+10.5} = \frac{1}{-7-10}$$

$$\Rightarrow \frac{I_1}{-10.5} = \frac{I_2}{9.5} = \frac{1}{-17}$$

$$\Rightarrow I_1 = \frac{-10.5}{-17} \quad \Big| \quad I_2 = \frac{9.5}{-17}$$

$$\therefore I_1 = 0.62 \quad \Big| \quad I_2 = -0.26$$

Fooom eqⁿ (iii), we get

$$I_3 = I_1 + I_2 = 0.62 - 0.26$$

$$\therefore I_3 = 0.36.$$

Therefore,

$I_1 = 0.62$

$I_2 = -0.26$

$I_3 = 0.36$

} Ans.

Section → B

← (4) →

Date: 11.10.21

Date: 11.01.22

Lecture: 17

① Faradays Experiment

⑪ Faradays law

⑪ self Inductance

⑫ Mutual Inductance

⑤ self Inductance for a long solenoid

⑥ Mutual inductance two coaxial coil.

Answer:

① Faradays Experiment:

